

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/Toyota.csv")
print(df)
```

	Unnamed: 0	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	\
0	0	13500	23.0	46986	Diesel	90	1.0	0	2000	
1	1	13750	23.0	72937	Diesel	90	1.0	0	2000	
2	2	13950	24.0	41711	Diesel	90	NaN	0	2000	
3	3	14950	26.0	48000	Diesel	90	0.0	0	2000	
4	4	13750	30.0	38500	Diesel	90	0.0	0	2000	
...	...	...	...	...	...	...	...	...	...	...
1431	1431	7500	NaN	20544	Petrol	86	1.0	0	1300	
1432	1432	10845	72.0	??	Petrol	86	0.0	0	1300	
1433	1433	8500	NaN	17016	Petrol	86	0.0	0	1300	
1434	1434	7250	70.0	??	NaN	86	1.0	0	1300	
1435	1435	6950	76.0	1	Petrol	110	0.0	0	1600	

	Doors	Weight
0	three	1165
1	3	1165
2	3	1165
3	3	1165
4	3	1170
...	...	...
1431	3	1025
1432	3	1015
1433	3	1015
1434	3	1015
1435	5	1114

[1436 rows x 11 columns]

```
df.head()
```

	Unnamed: 0	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors	Weight
0	0	13500	23.0	46986	Diesel	90	1.0	0	2000	three	1165
1	1	13750	23.0	72937	Diesel	90	1.0	0	2000	3	1165
2	2	13950	24.0	41711	Diesel	90	NaN	0	2000	3	1165
3	3	14950	26.0	48000	Diesel	90	0.0	0	2000	3	1165
4	4	13750	30.0	38500	Diesel	90	0.0	0	2000	3	1170

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.isnull().sum()
```

	0
Unnamed: 0	0
Price	0
Age	100
KM	0
FuelType	100
HP	0
MetColor	150
Automatic	0
CC	0
Doors	0
Weight	0
dtype:	int64

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1436 entries, 0 to 1435
Data columns (total 11 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Unnamed: 0   1436 non-null   int64
1   Price        1436 non-null   int64
2   Age          1336 non-null   float64
3   KM           1436 non-null   object
4   FuelType     1336 non-null   object
5   HP           1436 non-null   object
6   MetColor     1286 non-null   float64
7   Automatic    1436 non-null   int64
8   CC           1436 non-null   int64
9   Doors        1436 non-null   object
10  Weight       1436 non-null   int64
dtypes: float64(2), int64(5), object(4)
memory usage: 123.5+ KB
```

df.describe()

	Unnamed: 0	Price	Age	MetColor	Automatic	CC	Weight	
count	1436.000000	1436.000000	1336.000000	1286.000000	1436.000000	1436.000000	1436.000000	
mean	717.500000	10730.824513	55.672156	0.674961	0.055710	1566.827994	1072.45961	
std	414.681806	3626.964585	18.589804	0.468572	0.229441	187.182436	52.64112	
min	0.000000	4350.000000	1.000000	0.000000	0.000000	1300.000000	1000.00000	
25%	358.750000	8450.000000	43.000000	0.000000	0.000000	1400.000000	1040.00000	
50%	717.500000	9900.000000	60.000000	1.000000	0.000000	1600.000000	1070.00000	
75%	1076.250000	11950.000000	70.000000	1.000000	0.000000	1600.000000	1085.00000	
max	1435.000000	32500.000000	80.000000	1.000000	1.000000	2000.000000	1615.00000	

df.dtypes

```

      0
Unnamed: 0    int64
Price         int64
Age          float64
KM            object
FuelType      object
HP            object
MetColor     float64
Automatic     int64
CC            int64
Doors        object
Weight       int64

dtype: object
```

df.dropna()

	Unnamed: 0	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors	Weight
0	0	13500	23.0	46986	Diesel	90	1.0	0	2000	three	1165
1	1	13750	23.0	72937	Diesel	90	1.0	0	2000	3	1165
3	3	14950	26.0	48000	Diesel	90	0.0	0	2000	3	1165
4	4	13750	30.0	38500	Diesel	90	0.0	0	2000	3	1170
5	5	12950	32.0	61000	Diesel	90	0.0	0	2000	3	1170
...	...	...	...	...	...	...	...	...	...	...	...
1425	1425	7950	80.0	??	Petrol	86	1.0	0	1300	4	1000
1429	1429	8950	78.0	24000	Petrol	86	1.0	1	1300	5	1065
1430	1430	8450	80.0	23000	Petrol	86	0.0	0	1300	3	1015
1432	1432	10845	72.0	??	Petrol	86	0.0	0	1300	3	1015
1435	1435	6950	76.0	1	Petrol	110	0.0	0	1600	5	1114

1111 rows × 11 columns

```
mean_val=df['Age'].mean()
df['Age']=df['Age'].fillna(mean_val)
mean_val=df['MetColor'].mean()
df['MetColor']=df['MetColor'].fillna(mean_val)
df.isnull().sum()
```

	0
Unnamed: 0	0
Price	0
Age	0
KM	0
FuelType	100
HP	0
MetColor	0
Automatic	0
CC	0
Doors	0
Weight	0

dtype: int64

```
df['Price']=df['Price'].astype('float')
df.dtypes
```

```
0
Unnamed: 0    int64
Price         float64
Age           float64
KM            object
FuelType      object
HP            object
MetColor      float64
Automatic     int64
CC            int64
Doors         object
Weight        int64
```

dtype: object

df.head(5)

	Unnamed: 0	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors	Weight	
0	0	13500.0	23.0	46986	Diesel	90	1.000000	0	2000	three	1165	
1	1	13750.0	23.0	72937	Diesel	90	1.000000	0	2000	3	1165	
2	2	13950.0	24.0	41711	Diesel	90	0.674961	0	2000	3	1165	
3	3	14950.0	26.0	48000	Diesel	90	0.000000	0	2000	3	1165	